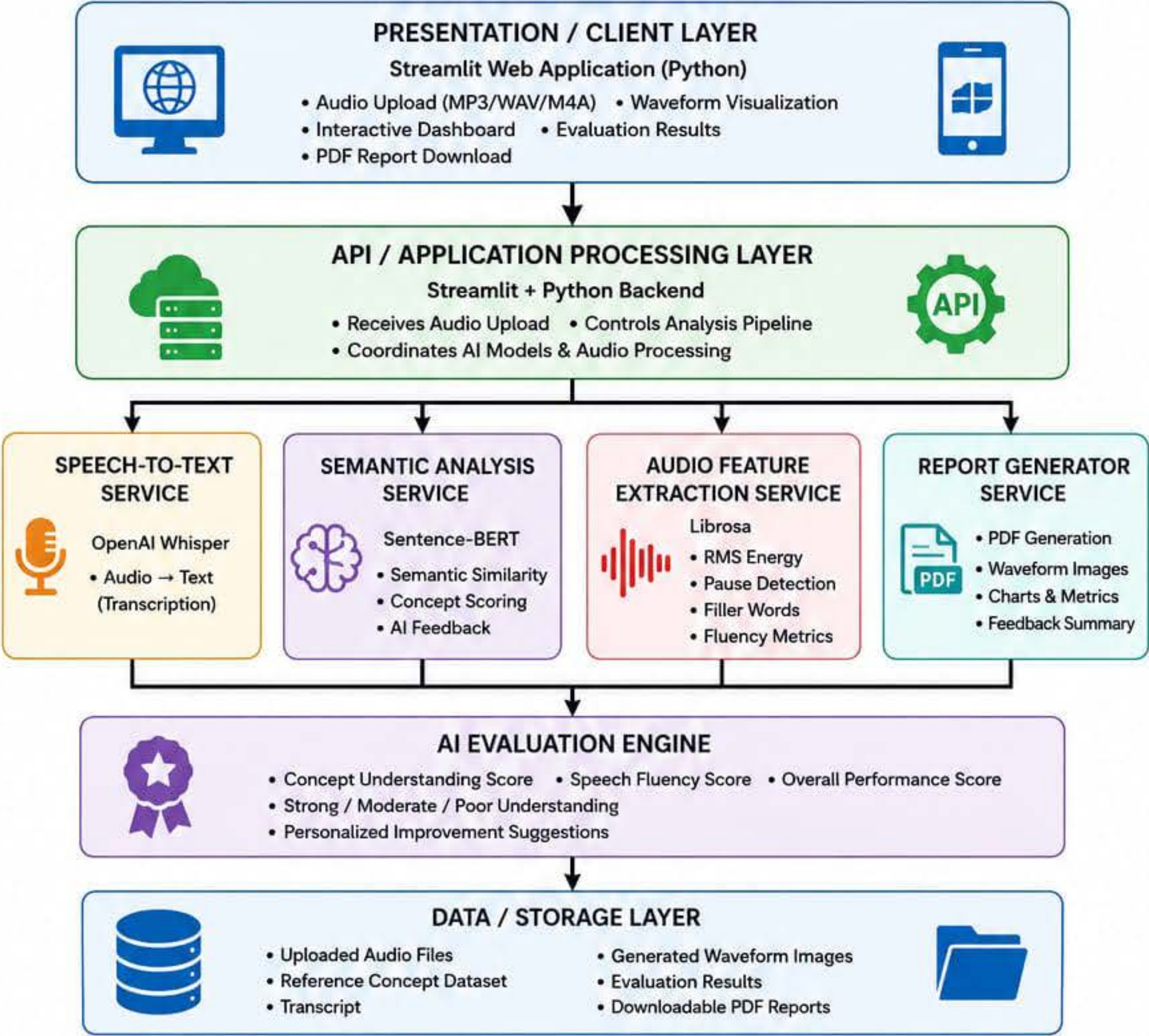


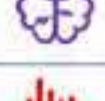
# SOLUTION ARCHITECTURE

|               |                                                    |
|---------------|----------------------------------------------------|
| Date          | 12/07/2026                                         |
| Name          | Karuru Arjun                                       |
| Project Name  | Voice-Based Concept Understanding Analyser (VBCUA) |
| Maximum Marks | 5 Marks                                            |

## SOLUTION ARCHITECTURE DIAGRAM



## COMPONENT DESCRIPTION TABLE

| Component                                                                                                            | Description / Role                                                     | Technologies Used            |
|----------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|------------------------------|
|  Presentation Layer               | User interface for audio upload, dashboard, visualization and reports. | Streamlit, HTML, CSS         |
|  Application Layer                | Controls workflow and coordinates AI modules.                          | Python, Streamlit            |
|  Speech-to-Text Service           | Converts speech into accurate text.                                    | OpenAI Whisper               |
|  Semantic Analysis Service        | Evaluates conceptual understanding using semantic similarity.          | Sentence-BERT, Transformers  |
|  Audio Feature Extraction Service | Extracts audio features for fluency analysis.                          | Librosa, NumPy               |
|  Report Generation Service        | Generates structured PDF reports with charts, waveform and feedback.   | FPDF / ReportLab, Matplotlib |
|  AI Evaluation Engine             | Calculates scores and provides qualitative feedback & suggestions.     | Python, NLP Models           |
|  Data / Storage Layer             | Stores audio files, datasets, transcripts, results and PDF reports.    | File System, CSV, SQLite     |